

# Manipulate to Obfuscate: A Privacy-Focused Intelligent Image Manipulation Tool for End-Users

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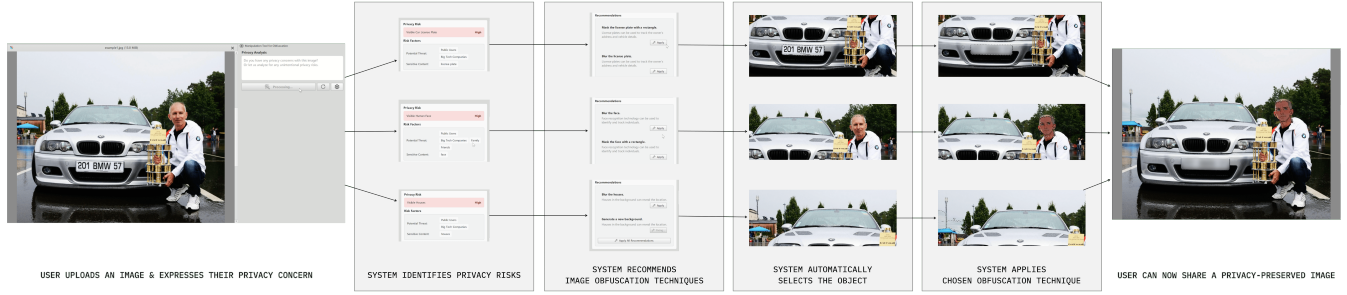


Figure 1: Manipulate to Obfuscate helps users identify and mitigate privacy risks in an image.

## ABSTRACT

Image-related privacy preservation techniques often demand significant technical expertise, creating a barrier for end-users. We present a privacy-focused intelligent image manipulation tool that leverages recent advancements in generative AI to lower this barrier. Our functional prototype allows users to express their privacy concerns, identify potential privacy risks in images, and recommends relevant AI-powered obfuscation techniques to mitigate these risks and concerns. We demonstrate the tool’s versatility across multiple different domains, showcasing its potential to empower users in managing their privacy across various contexts. This demonstration presents the concept, user workflow, and implementation details of our prototype, highlighting its potential to bridge the gap between privacy research and practical, user-facing tools for privacy-preserving image sharing.

## CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**; • **Security and privacy** → **Usability in security and privacy**.

## KEYWORDS

image privacy, usable security, generative artificial intelligence

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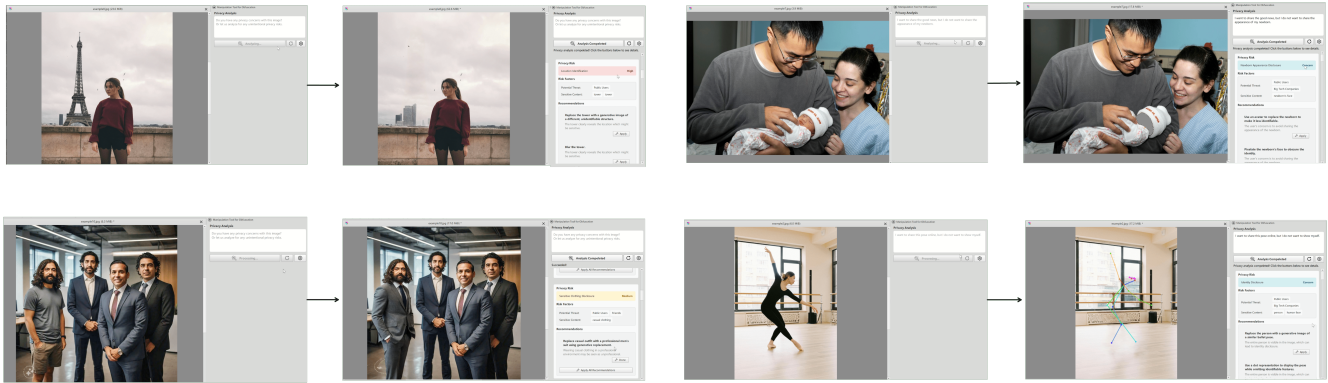
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## 1 INTRODUCTION

Image sharing has become an integral part of our digital lives, driven by the proliferation of smartphones with high-quality cameras, increasing the importance of visual communication and social media platforms. These images, rich in information, can inadvertently reveal more than intended, posing significant privacy risks to users, like self disclosure and location exposure [1, 13]. To help users identify and mitigate these risks, prior work has explored avenues of research, e.g., categorizing sensitive content in images [10, 11], developing tools for automatic identification of sensitive elements [2, 3, 10], developing and assessing the effectiveness of image obfuscation techniques [5, 6, 9, 17], and taxonomizing risks and defenses to further enhance our understanding of image privacy more generally [4, 7, 12]. However, these valuable insights are limited to academia and deploying these techniques often requires significant technical expertise, creating a significant barrier for end-user use.

We lower this barrier by creating an intelligent image manipulation tool — “Manipulate to Obfuscate” — that allows users to: articulate privacy concerns with an image, automatically identify pertinent risks in those images relevant to those concerns, and then mitigate those risks using a panoply of relevant obfuscation techniques. Our functional prototype combines a Multimodal Large Language Model (MLLM) to identify sensitive elements in an image with AI-powered image manipulation. To help users identify pertinent risks, we use few-shot learning techniques drawing on image privacy risks from known taxonomies proposed in prior work. To



**Figure 2: The tool helps mitigate various privacy risks, including location disclosure, identity exposure, and user-specific privacy concerns. It recommends and applies obfuscation techniques such as blurring, inpainting, and generative replacement.**

mitigate these risks, we recommend different image obfuscation techniques informed by known taxonomies proposed in prior work. We automate the application of the image obfuscation technique by leveraging computer vision and AI image generation models and an open-source graphics editing tool. The main contribution of this work is presenting the concept, user workflow, and implementation details of a privacy-focused intelligent image obfuscation tool.

## 2 MANIPULATE TO OBFUSCATE

“Manipulate to Obfuscate” leverages generative AI tools like MLLMs and diffusion-powered AI image generators to help users apply privacy-preserving image obfuscation techniques to their images.

### 2.1 User Workflow

- (1) **Image Upload and Concern Expression:** Users upload their image content to the system, initiating the privacy risk assessment process. The tool asks the user to express any privacy concerns they have with sharing their image online.
- (2) **Privacy Risk Assessment:** Based on the image and the privacy concern mentioned, the system analyzes for potential unintended privacy risks and sensitive information. The system presents a comprehensive privacy analysis to the user, detailing identified risks and their severity.
- (3) **Image Obfuscation Techniques:** Based on each privacy concern mentioned by the user and the potential privacy risk and sensitive content identified by the system, the system recommends mitigation strategies in the form of obfuscation techniques.

### 2.2 Implementation

The implementation has three major components - prompting technique, graphic editor interface and tools, and image obfuscation techniques.

**2.2.1 Prompting Approach.** The system uses OpenAI’s GPT-4o model as its Multimodal Large Language Model (MLLM) to analyze image content and identify privacy risks. Using chain-of-thought prompting [15], the MLLM performs a comprehensive analysis:

**Sensitive Content Detection:** The MLLM identifies sensitive content based on the taxonomy proposed by Li et al. [11]. Additionally, it is prompted to adapt and recognize potential sensitive elements beyond this predefined list.

**Privacy Risk Assessment:** The system evaluates sensitive content to determine potential privacy risks, contextualizing them in real-world scenarios.

**Obfuscation Recommendations:** To mitigate privacy risks, the system recommends obfuscation techniques including blurring, pixelation, silhouette, avatar, point-light, bar, masking, in-painting, and generative content replacement, informed by literature taxonomies [4, 7, 12].

**2.2.2 Graphic Editor Interface and Tools.** Our prototype is built on top of Krita, an open-source raster graphics editor [14]. The key modifications over the naive graphic editor include:

**Risk Presentations:** The interface has a sidebar that presents a ranked list of identified privacy risks, prioritized by their severity scores, and composed of actionable recommendations.

**Sensitive Element Precise Selection:** We use the Florence vision model to locate sensitive elements and obtain their bounding boxes [16], then apply the Segment Anything Model for precise selections of the elements [8].

**Interactive Highlighting:** As users review the risk recommendations, the system highlights corresponding sensitive elements in the image when users hover over specific recommendations.

**2.2.3 Image Obfuscation Techniques.** Our system implements a comprehensive suite of obfuscation techniques to address various privacy concerns:

**Traditional Methods:** Blurring, Pixelation, Silhouette, and Masking were implemented using our Sensitive Element Precise Selection tool combined with Krita’s built-in features.

**AI-Powered Techniques:** Avatar Replacement, In-painting, Point-Light Representation, Bar Representation, and Generative Content Replacement leverage diffusion models for realistic and context-aware obfuscation.

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